

# 1 Overall structure

We are trying to make use of a Levins-style metapopulation model to explain the reduction of virulence in bubonic plague. We hope to show that strains with a lower virulence would be favored at metapopulation level.

If every individual in a patch is not infected, we say this patch is "susceptible", and if there is any infected individual in the patch we say it is "infected".

Let  $p_I = p$  denote fraction of infected patches and fraction of uninfected patches is  $p_S = 1 - p$ .

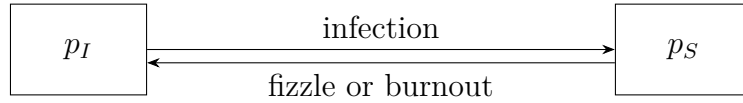


Figure 1: patch state transition

Susceptible patches can be infected because of host movement (infected individuals from infected patches move into susceptible patches or contact with individuals in susceptible patches and infect them) .

Infected patches can become susceptible again because of fizzle or burnout of the within-patch epidemic.

We also take population density of patches into account because burnout probability is a function of population size.

## 2 Models

### 2.1 three-dimensional model

In this model we keep track of three state variables: fraction of infected patches  $p$ , average population density in infected patches  $N_I$ , and average population density in susceptible patches  $N_S$ .

We have several events that can impact system dynamics:

- **migration:**

$$\begin{aligned}
 N_S &\rightarrow N_S - cp(N_S - N_I) \\
 N_I &\rightarrow N_I + c(1 - p)(N_S - N_I)
 \end{aligned}$$

Suppose individual rats move with probability  $c$ , so total number of rats migrating is  $c((1 - p)N_S + pN_I)$ , and they randomly land in patches with equal probability. So  $cN_S$  rats leaves a susceptible patch and  $c((1 - p)N_S + pN_I)$  lands in it, and the overall change in susceptible patches is  $c((1 - p)N_S + pN_I) - cN_S = cp(N_S - N_I)$ , and similarly for infected patches.

- **infection:**  $p \rightarrow p + (1 - e^{-Bp})(1 - p)$

We use hazard model to describe the amount of newly infected patches. We are currently assuming constant  $B$ , and can come back to this point and consider more realistic assumptions (for example,  $B$  increases with  $N_I$  and  $N_S$ ).

- **fizzle or burnout:**  $p \rightarrow P_1 p$

where  $P_1 = P_1(R_0, \varepsilon, k = 1, N_I)$  is persistence probability.

- **epidemic:**  $N_I \leftarrow (1 - zD)N_I$

where  $z = z(R_0)$  is the final size and  $D$  is the probability that an infected host dies from pathogen.

- **growth:**

$$\begin{aligned} N_S &\leftarrow N_S + rN_S(1 - \frac{N_S}{K}) \\ N_I &\leftarrow N_I + rN_I(1 - \frac{N_I}{K}) \end{aligned}$$

- **change of average population size when some patch state changes:** calculate the average again after change of patch state. That is,

when  $\Delta p > 0$ ,

$$N_I \leftarrow \frac{pN_I + \Delta p N_S}{p + \Delta p}$$

and  $N_S$  does not change.

when  $\Delta p < 0$

$$N_S \leftarrow \frac{(1 - p)N_S - \Delta p N_I}{1 - p - \Delta p}$$

and  $N_S$  does not change.

We denote infection event with  $f_i$ , epidemic and burnout with  $f_e$ , movement with  $f_m$ , growth with  $f_g$ , then we have

$$\begin{aligned} f_m(p, N_I, N_S) &= \begin{pmatrix} p \\ N_I + c(1 - p)(N_S - N_I) \\ N_S - cp(N_S - N_I) \end{pmatrix} \\ f_i(p, N_I, N_S) &= \begin{pmatrix} p + Bp(1 - p) \\ N_I + B(1 - p)N_S \\ 1 + B(1 - p) \\ N_S \end{pmatrix} \\ f_e(p, N_I, N_S) &= \begin{pmatrix} P_1(N_I)p \\ (1 - zD)N_I \\ (1 - p)N_S + (1 - zD)(1 - P_1)pN_I \\ 1 - P_1p \end{pmatrix} \end{aligned}$$

$$f_g(p, N_I, N_S) = \begin{pmatrix} p \\ N_I + rN_I \left(1 - \frac{N_I}{K}\right) \\ N_S + rN_S \left(1 - \frac{N_S}{K}\right) \end{pmatrix}$$

$$(p_{t+1}, N_{I_{t+1}}, N_{S_{t+1}}) = f(p_t, N_{I_t}, N_{S_t})$$

$$f = f_g \circ f_e \circ f_i \circ f_m$$

However the equations are too complicated for further analysis, so we try to simplify it.

## 2.2 assuming fixed population density in susceptible patches

Simulation of the three-dimensional model indicates that population density of susceptible patches doesn't change a lot over time, so we simplify the model by assuming that  $N_S$  is constant. This leads to a two-dimensional model with state variables  $p$  and  $N_I$ .

### 2.2.1 burnout before spread

Let  $p$  denote fraction of infected patches and  $N$  denote population density of infected patches.

**epidemic:** Infected hosts die and so some infected patches burnout.

$$f_e : \begin{pmatrix} p \\ N \end{pmatrix} \rightarrow \begin{pmatrix} P_1 p \\ (1 - zD)N \end{pmatrix}$$

$P_1 = \text{P1\_prob}(N)$  is the persistence probability,  $z$  is final size of this epidemic wave,  $D$  is the death probability of infected hosts.

**growth:** Assume a constant birth rate  $r$ .

$$f_g : \begin{pmatrix} p \\ N \end{pmatrix} \rightarrow \begin{pmatrix} p \\ (1 + r)N \end{pmatrix}$$

$1 + r$  multiplier is chosen in infected patches just for math simplicity since plague should kill a large proportion of hosts so it would be far from carrying capacity. In susceptible patches we still have a finite carrying capacity so the population is bounded.

**spread:** Hosts move around and simultaneously bring pathogens to susceptible patches (combining movement and infection in one step for concision).

$$f_s : \begin{pmatrix} p \\ N \end{pmatrix} \rightarrow \begin{pmatrix} p + (1 - e^{-Bp})(1 - p) \\ N + c(1 - p)(S - N) \end{pmatrix}$$

$B$  is infection rate,  $c$  is moving probability of individual rats, and  $S$  is the population density of susceptible patches (assumed constant).

The overall iteration process is

$$f = f_s \circ f_g \circ f_e : \begin{pmatrix} p \\ N \end{pmatrix} \rightarrow \begin{pmatrix} P_1(p + (1 - e^{-Bp})(1 - p)) \\ N(1 + r)(1 - zD) + c(1 - p)(S - N(1 + r)(1 - zD)) \end{pmatrix}$$

**Disease Free Equilibrium:**

If we start at a point where  $p > 0, N > 0$ , all variables remain positive during the iterations as each step maps positive values to a positive values.

If we start at  $p = 0, N > 0$  then population size ends up at

$$N_t = N_{t+1}$$

i.e.

$$N_{DFE} = \frac{cS}{1 - (1+r)(1-zD)(1-c)}$$

$N_{DFE}$  increases with  $r, c, S$  and decreases with  $z, D$ .

**Endemic Equilibrium:**

We have an endemic equilibrium but don't have an analytical solution. And we haven't prove that EE is stable)

**invasion analysis**

$$J_{f_e} = \begin{pmatrix} P_1 & p \frac{\partial P_1}{\partial N} \\ 0 & 1 - zD \end{pmatrix}$$

$$J_{f_g} = \begin{pmatrix} 1 & 0 \\ 0 & 1 + r \end{pmatrix}$$

$$J_{f_s} = \begin{pmatrix} (1 + (1-p)B)e^{-Bp} & 0 \\ -c(S - N) & 1 \end{pmatrix}$$

And at DFE we have  $p = 0, N = N_{DFE}$  so the matrices above are all lower triangular, so eigenvalues of  $J_f = J_{f_s} J_{f_g} J_{f_e}$  are  $\lambda_1 = (1+B)P_1$  and  $\lambda_2 = (1-zD)(1+r)$ . We required previously that  $\lambda_2$  is always smaller than 1. If  $(1+B)P_1|_{N=N_{DFE}} = (1+B)P_1|_{N=\frac{cS}{1-(1+r)(1-zD)(1-c)}} < 1$ , DFE is stable, so the disease cannot persist; if it is greater than 1, DFE is unstable. Simulation shows that the disease will persist and system ends up at endemic equilibrium.

We define the first eigenvalue as landscape reproduction number  $L_0 = (1+B)P_1|_{N=\frac{cS}{1-(1+r)(1-zD)(1-c)}}$ , so the criterion for plague persistence is  $L_0 > 1$ .

$P_1$  is non-monotonous with  $R_0$ , but in the decreasing region,  $P_1$  is too low (about  $10^{-0.5}$   $10^{-12}$ ) so  $B$  has to be very high to make it persist, which seems unlikely.

Another possibility is that decreased mortality  $D$  makes  $P_1$  higher to destabilizes DFE and make plague persist. Figures in the repo shows the region where this might can happen when  $D$  decreases from 1 to 0.85.

**2.2.2 spread before burnout**

In previous models, burned-out patches are set to be susceptible in the next season and have no chance to spread disease. But it is very likely that a patch with a large epidemic can infect

other patches before it burns out.

At the beginning of the season if the proportion of infected patches is  $p$  and population density is  $N$ , then at the beginning of next season  $p$  becomes  $P_1p + (1 - e^{-Bp})(1 - p)$  instead of the previous  $P_1(p + (1 - e^{-Bp})(1 - p))$

The change of  $N$  is the same as previous version:  $N \rightarrow (1 + r)(1 - zD)N + c(1 - p)(S - (1 + r)(1 - zD)N)$ .

So the overall function is

$$f : \begin{pmatrix} p \\ N \end{pmatrix} \rightarrow \begin{pmatrix} P_1p + (1 - e^{-Bp})(1 - p) \\ (1 + r)(1 - zD)N + c(1 - p)(S - (1 + r)(1 - zD)N) \end{pmatrix}$$

The disease-free equilibrium is  $p = 0$  and

$$N_{DFE} = \frac{cS}{1 - (1 - c)(1 + r)(1 - zD)}$$

Local linearization at DFE is

$$J_f = \begin{pmatrix} P_1|_{N=\frac{cS}{1-(1-c)(1+r)(1-zD)}} + B & 0 \\ -c(S - (1 + r)(1 - zD)N) & (1 - c)(1 + r)(1 - zD) \end{pmatrix}$$

So the persistence criterion is

$$L_0 > 1$$

where

$$L_0 = P_1|_{N=\frac{cS}{1-(1-c)(1+r)(1-zD)}} + B$$

$B$  is a constant, so  $L_0$  is non-monotonous with  $R_0$  and the decreasing interval is the same with  $P_1$ . It can happen that at the beginning,  $L_0$  is slightly greater than 1, but there is a decrease in  $B$  for some reason and makes  $L_0 < 1$ . Then decreasing  $R_0$  (or decreasing  $D$ ) makes  $P_1$  larger and  $L_0 > 1$  again.

### 2.2.3 transmission proportional to size of within-patch epidemic

Specially, following the previous version, we can further assume that the transmission for next season is proportional to the size of the epidemic in a patch, i.e., let  $B = azP_{nf}$ , where  $a$  is a scaling constant,  $P_{nf} = 1 - \frac{1}{R_0}$  is the non-fizzle probability and  $z$  is the final size. So we have

$$L_0 = P_1|_{N=\frac{cS}{1-(1-c)(1+r)(1-zD)}} + azP_{nf}$$

$P_1$  is non-monotonous with  $R_0$  but  $zP_{nf}$  is increasing, so it can be harder to find the scenario we want. When  $a$  is high  $L_0$  is dominant by  $zP_{nf}$ , so it's likely to be increasing, but when  $a$  is low  $L_0$  is far from 1.

We might consider alternatives that  $B$  is still increasing with  $R_0$  but slower than this one, for instance,  $B = az^\gamma P_{nf}$ , where  $0 < \gamma < 1$ . The problem is there doesn't seem to be proper data to estimate new parameters we introduced.

## 2.3 averaging population density of all patches

In previous version  $N_{DFE}$  is a function of many parameters  $(c, r, S, R_0, D)$  and moving probability  $c$  might be difficult to estimate. And if a patch gets infected and then burns out, its population size should be  $(1+zD)N$ , but our assumption makes it suddenly go up immediately to  $S$ .

To solve these we consider modeling average population density of all patches instead of infected patches only, so moving probability  $c$  is no longer necessary, and it's changing continuously, which makes things easier and more reasonable.

We assume that the population growth follows a logistic model:

$$N \rightarrow N(1 + r - \frac{rN}{K})$$

And a proportion of  $zDP_{nf}p$  of the population dies because of the epidemic,  $P_{nf}$  is the non-fizzle probability.

So the overall function is

$$f : \begin{pmatrix} p \\ N \end{pmatrix} \rightarrow \begin{pmatrix} P_1p + (1 - e^{-Bp})(1 - p) \\ (1 - zDP_{nf}p)(1 + r - \frac{rN}{K})N \end{pmatrix}$$

The disease-free equilibrium is  $p = 0$  and

$$N_{DFE} = K$$

Local linearization at DFE is

$$J_f = \begin{pmatrix} B + P_1|_{N=K} & 0 \\ -zDP_{nf} & 1 - r \end{pmatrix}$$

So the persistence criterion is

$$L_0 > 1$$

where

$$L_0 = P_1|_{N=K} + B$$

This also works since  $P_1$  is still non-monotonous with  $R_0$ .

## 3 Parameters

In models with logistic growth, we want to constraint the growth rate  $r$  in  $(0, 1)$ . In models with a simple exponential growth,  $r$  can be anything, but we need  $(1+r)(1-zD) < 1$  to make sure the second eigenvalue of Jacobian matrix at DFE is smaller than 1 (otherwise it means DFE is always unstable and disease always persists).

Moving probability  $c$  and death probability  $D$  should all be in  $(0, 1)$ . Researches show that  $D = 1$  for wild-type strains and  $D = 0.85$  for pla-reduced strains.

$\varepsilon$  is the mean infectious period as a fraction of host life time/generation interval. References indicate mean infectious period is about 7.4–9.2 days, but we need more discussion on rat generation interval.

There are not any references giving  $R_0$  of bubonic plague in rats but some gives epidemic doubling-time  $T_d = 3.714$  in humans. If we assume  $T_d$  is the same for rats, and we can use the same formula  $R_0 = 1 + \frac{T}{T_d} \ln 2$  (where  $T$  is the infectious period of rats) to estimate  $R_0$ . If we put  $T = 8$  into the formula we get  $R_0 = 2.5$

## 4 next step

We might discuss if current models are reasonable, and if so, we can try to prove that the pla-reduced strains can invade wild-type strains.

## 5 points to reconsider/refine

- We are assuming there is always only 1 initially infected individual in an infected patch
- We are dealing with average population density in patches. This means that we are neglecting variation in population density within each group.
- We are assuming that the population density of susceptible patches is constant (except the first and last one).
- Burnout probability is derived based on the assumption that birth rate equals natural death rate, and all infected individuals will recover and won't die because of disease.
- We are assuming that  $R_0$  is independent of host population size  $N$  (frequency-dependent transmission). We might consider  $R_0$  scaling linearly with  $N$  (density-dependent transmission) or something between.
- We are assuming that a burned-out patch is susceptible in the next season as if it was never infected, and a patch that persists the first wave can still go extinct in the next wave like a newly infected patch.